

Course Outline

Course Code, Number, and Title:

DANA 4840 SECTION 001 CLASSIFICATION II

Year and semester: Summer 2026

Course Format: Lecture 4.0h

Credits: 3.0

Course Description, Prerequisites, and Corequisites:

Course Description: Following dimension reduction and standard techniques of classification, situations arise where more advanced techniques are called for. Students learn the various multivariate techniques for classifying objects or cases into several groups. Density-based and centroid-based clustering, hierarchical techniques, as well as other clustering techniques such as fuzzy clustering will be discussed in detail.

Learning Outcomes:

Upon successful completion of this course, students will be able to:

- Identify the different types of clustering techniques
- Use SAS or R to implement hierarchical clustering techniques
- Use SAS or R to implement density-based clustering techniques
- Use SAS or R to implement centroid-based clustering techniques
- Use SAS or R to implement other clustering techniques
- Review results of a clustering process
- Evaluate and assess result of a clustering process

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Instructor(s): Yew-Wei Lim

Office: B153a

Email: ywlim@langara.ca

Office Hours: Mon and Wed: 5:30 pm - 6:20 pm

Tue and Thu: 1:30 pm - 2:20 pm

Textbook and Course Materials:

The following textbook is required for this course:

- Practical Guide to Cluster Analysis in R, by Alboukadel Kassambara, sthda.com

Assessments and Weighting:

	Percentages	Dates
Final Examination	35%	04 – 15 August
Midterms (2)	30%	MT1: 15 June; MT2: 20 Jul
Term Project	25%	TBA
Assignments/Class Participation	10%	TBA

Grading:

Grades, Grade Points, & Notations

The following grading scheme is used in this course:

Minimum Overall (%)	90	85	80	76	72	68	64	60	55	50
Grade in Course	A+	A	A-	B+	B	B-	C+	C	C-	D

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Detailed Course Schedule:

Unit	Topic	Textbook Reference
1	Clustering distance measures	Chapter 3
2	Partitioning clustering	Chapter 4-7
3	Hierarchical clustering	Chapter 8-13
4	Cluster validation	Chapter 14-19
5	Advanced clustering techniques	Chapter 20-24 (optional topics)

Policy

As a student at Langara, you are responsible for familiarizing yourself and complying with the following policies:

College Policies:

- Student Code of Conduct - E1003
- Academic Integrity - F1004
- Academic Standing - Academic Probation and Academic Suspension - E2008
- Appeal of Final Grade - E2006
- Concerns about Instruction - F1002
- Withdrawal from Courses and Deferred Standing - E2011

Departmental/Course Policies:

1. The final examination will be a three-hour exam on the entire semester's work.
2. The final examination period for this term is 04 – 15 August, 2026. Seating at the final exam is mandatory. Hence, students should make themselves available during this time.
3. The last day of withdrawal this term is Saturday June 27, 2026.
4. Students must have a calculator with the “square root” function ($\sqrt{x}$).
5. No make-up test will be given. If you miss a test because of personal or medical reasons, please contact your instructor via email ([mail to: ywlim@langara.ca](mailto:ywlim@langara.ca)) as soon as possible.

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6. Please be prepared to spend at least an hour each day outside of class time to read the textbook, work on assignments and prepare for tests/exam.
7. The mid-term tests and final examination are cumulative, testing you on all topics covered throughout the term.
8. Students should be familiar with their responsibilities and avoid plagiarism, including avoiding the use of a solution provider “service” such as Chegg Study or having another person help with writing an assignment.
9. If you make use of AI (ChatGPT and other artificial intelligence services) for any portion of your assignments/projects to obtain marks for credit, you are required to seek permission from your instructor beforehand.
10. Students will have to be physically present for all in-person activities/exams and be prepared to show proof of their identification.

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